

Feedback Ordering Effects in LLM Code Repair

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Abstract

LLM-based code repair systems iteratively refine generated code using feedback from test execution and static analysis. While prior work has studied which feedback types improve repair quality, the effect of feedback *ordering* under matched content remains unexplored. We hypothesize that presenting static analysis errors before execution failures creates a coarse-to-fine repair trajectory that improves outcomes. Our experiments on HumanEval and MBPP with GPT-4o-mini confirm this: static→execution ordering achieves 29% relative improvement over execution→static ordering, with byte-identical feedback content in both conditions. Investigating the mechanism, we find the effect operates through regression prevention—static-first repairs exhibit 38% fewer inter-iteration regressions—rather than early-gain acceleration. These findings suggest that feedback structure, not just content, is a first-order design variable in LLM repair systems, offering a free performance gain to practitioners who reorder existing feedback pipelines.

1. Introduction

When debugging code, do you fix typos before logic errors, or dive straight into the failing test? For humans, the answer is instinctive—clean up syntax first, then tackle semantics. Yet LLM-based code repair systems treat all feedback equally, presenting execution failures and static analysis warnings in arbitrary order. This seemingly trivial presentation choice, we show, causes a 29% performance difference in repair quality.

LLM self-repair has emerged as a promising approach to code generation, achieving 10–17% relative improvement on standard benchmarks (Olausson et al., 2024). The dominant paradigm iteratively refines generated code using exe-

cution feedback—test results, stack traces, and assertion failures—until tests pass or a budget is exhausted. Recent work has explored augmenting execution feedback with static analysis (type checking, linting, style violations), demonstrating significant reductions in security and reliability issues (Blyth et al., 2025).

However, existing studies conflate feedback *content* with feedback *presentation*. When comparing “static analysis feedback” against “execution feedback,” prior work necessarily varies the information provided—more feedback types mean more information. This leaves a fundamental question unanswered: does the **order** in which feedback is presented independently affect repair quality, holding content constant?

This gap matters because presentation order is a free variable in any feedback-based repair system. If order affects quality—and if that effect is large—then current systems leave significant performance on the table by ignoring this structural choice. Moreover, understanding *why* order matters would reveal insights into how LLMs process sequential feedback during iterative repair.

We hypothesize that static-first feedback ordering creates a coarse-to-fine repair trajectory. When static analysis errors (syntax violations, type mismatches, style warnings) appear before execution failures, the LLM resolves surface-level issues before tackling semantic errors. This clears low-level noise from the repair space, providing a cleaner foundation for logical reasoning. The key insight is not that static-first provides different information—we control for that—but that it structures the repair process differently.

Our experiments confirm this hypothesis with surprising strength. On HumanEval (164 problems) + MBPP (500 problems) with GPT-4o-mini, static→execution ordering achieves 55.57% pass@1 versus 43.07% for execution→static—a **29.02% relative improvement** (95% CI: [15.74%, 44.53%], $p = 6.31 \times 10^{-6}$). Critically, both conditions receive byte-identical feedback content; only the presentation order differs.

Investigating the mechanism, we find that the effect operates through regression prevention rather than early-gain amplification. Static-first repairs exhibit 38% fewer inter-iteration regressions ($p = 0.0198$)—previously passing tests

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

stay passing. Early iteration gains, by contrast, are statistically indistinguishable between conditions. The scaffolding metaphor applies, but not in the way one might expect: static-first ordering doesn’t help the LLM build faster, it helps it build more stably.

We make the following contributions:

1. **Matched-content experimental design:** We introduce a methodology that isolates feedback ordering from information volume by ensuring byte-identical content across conditions. This controls the primary confound in prior feedback comparison studies.
2. **Evidence of ordering effect:** We demonstrate that static→execution ordering achieves 29% relative pass@1 improvement over execution→static ordering, with strong statistical significance ($p < 10^{-5}$).
3. **Mechanism identification:** We show that the effect operates through regression prevention (38% reduction) rather than early-gain acceleration, refining the scaffolding hypothesis.

The remainder of this paper is organized as follows. Section 2 reviews related work on LLM code repair and feedback mechanisms. Section 3 describes our experimental methodology. Section 4 presents the experimental setup, Section 5 reports results, and Section 6 discusses implications and limitations. Section 7 concludes with future directions.

2. Related Work

Our work bridges LLM code repair and feedback mechanism design. We review each area, highlighting the gap our matched-content design addresses.

2.1. LLM Self-Repair

Iterative self-repair has become the dominant paradigm for improving LLM code generation beyond single-shot accuracy. Olausson et al. (2024) demonstrated that feeding execution feedback (test results, stack traces) back to the model enables 12–17% relative improvement on HumanEval. Arimbur (2026) showed that most gains concentrate in the first 2–3 repair iterations, with diminishing returns thereafter.

Several frameworks operationalize self-repair. CodeRL (Le et al., 2022) integrates reinforcement learning with execution feedback. ThinkRepair (ISSTA 2024) uses self-directed debugging on Defects4J. More recent work explores multi-agent architectures: CodeCoR (2025) achieves 77.13% pass@1 on HumanEval/MBPP through agent collaboration.

These approaches share a common assumption: feedback content determines repair quality. The question of feedback *ordering*—given the same content—remains unexplored.

2.2. Static Analysis Feedback

Static analysis provides complementary signals to execution feedback. Jain et al. (2024) showed that LLM-assisted code cleaning improves code quality without runtime signals. Blyth et al. (2025) demonstrated that Pylint feedback reduces security issues from 40% to 13% and reliability warnings from 50% to 11%.

AutoSafeCoder (Nunez et al., 2024) combines static analysis with fuzz testing in a multi-agent framework, achieving 13% vulnerability reduction. Dolcetti et al. (2024) integrated testing and static analysis feedback for safety improvements.

However, studies comparing static and execution feedback face a fundamental confound: they necessarily vary information volume alongside feedback type. Cascaded approaches (static + execution) provide more information than single-type approaches. Our prior work (h-c1) achieved 16.18% improvement with cascaded feedback, but whether this stems from *more* information or *ordered* information was unclear.

2.3. Feedback Presentation Effects

Research on prompt engineering and positional effects suggests LLMs are sensitive to presentation order. Recency effects in transformer attention mean later tokens receive disproportionate weight. This raises an alternative hypothesis: static-first might help not because it structures the repair process, but simply because execution feedback appears last (terminal position bias).

FeedbackEval (Dai et al., 2025) benchmarked feedback-driven code repair across feedback types, finding mixed feedback yields 63.6% repair success. However, ordering effects were not systematically tested.

2.4. Our Contribution

Unlike prior work, we control for information volume by ensuring byte-identical feedback across conditions. This matched-content design isolates the ordering effect from the content effect. We test both the existence of an ordering effect and its mechanism—distinguishing scaffolding (coarse-to-fine repair) from recency bias (terminal position advantage).

3. Methodology

Our methodology isolates feedback ordering from information volume through a matched-content experimental design. We describe the design rationale, conditions, and evaluation

approach.

3.1. Design Rationale

Building on our observation that feedback ordering may independently affect repair quality, we design an experiment that controls the primary confound in prior studies: information volume. Rather than comparing “static feedback” vs “execution feedback” (which varies content), we compare “static→execution” vs “execution→static” (which varies only order).

Key Design Principle: Both conditions receive identical feedback content; only presentation order differs.

This requires:

1. Generating both static and execution feedback for each problem
2. Truncating each to a fixed token budget (500 tokens per type, 1000 total)
3. Concatenating in different orders for each condition
4. Ensuring byte-identical content across conditions

3.2. Experimental Conditions

We define two matched conditions:

Condition A (Static→Execution):

```
[Static Analysis Feedback: 500 tokens]
[Execution Feedback: 500 tokens]
```

Condition B (Execution→Static):

```
[Execution Feedback: 500 tokens]
[Static Analysis Feedback: 500 tokens]
```

The feedback content is generated once per problem-iteration pair, then presented in both orders. Deterministic truncation (first 500 tokens) ensures identical content.

3.3. Feedback Generation

Static Analysis: We run Pylint and Mypy on generated code. Pylint catches style violations, unused variables, and potential bugs. Mypy catches type errors when type hints are present. Output is concatenated and truncated to 500 tokens.

Execution Feedback: We execute code against test cases in a sandboxed environment, capturing stdout/stderr, pass/fail status, and stack traces for failures. Output is truncated to 500 tokens, preserving traceback structure.

3.4. Repair Loop

Both conditions use identical generation/execution logic; only concatenation order differs. The loop terminates early if all tests pass or after 3 iterations.

3.5. Evaluation Metrics

Primary Metric: pass@1—fraction of problems where final code passes all tests.

Relative Improvement: $(\text{pass@1}_A - \text{pass@1}_B) / \text{pass@1}_B \times 100\%$

Statistical Tests: Bootstrap CI (10,000 resamples) for 95% CI; McNemar’s test for paired comparison.

Mechanism Metrics: $\Delta\text{Pass}_{1 \rightarrow 2}$ (change in pass rate from iteration 1 to 2); $\text{RegRate}_{1 \rightarrow 2}$ ($P(\text{pass@iter1} \wedge \text{fail@iter2})$).

4. Experimental Setup

We design experiments to answer the following questions:

RQ1: Does feedback presentation order affect LLM code repair quality under matched content?

RQ2: If an ordering effect exists, what is its mechanism—early-gain amplification or regression prevention?

4.1. Datasets

HumanEval (Chen et al., 2021): 164 hand-crafted Python programming problems with comprehensive test suites.

MBPP (Austin et al., 2021): 500 crowd-sourced Python programming problems covering a broader difficulty range.

Table 1. Dataset statistics

Dataset	Problems	Avg. Tests/Problem	Difficulty
HumanEval	164	~5	Medium-Hard
MBPP	500	~3	Easy-Medium
Total	664	~4	Varied

4.2. Implementation Details

Model: GPT-4o-mini (temperature 0.0)

Token Budget: 500 + 500 (balanced, fits context)

Iterations: 3 (standard in literature)

Static Analysis: Pylint 3.0+, Mypy 1.0+

Execution: Sandboxed Python 3.10 subprocess, 10-second timeout

5. Results

We present evidence for the ordering effect (RQ1) and its mechanism (RQ2).

5.1. Main Results (RQ1)

Table 2. Main results on HumanEval + MBPP

Condition	pass@1	95% CI
Static→Execution (A)	55.57%	—
Execution→Static (B)	43.07%	—
Relative Improvement	29.02%	[15.74%, 44.53%]
McNemar p -value	6.31×10^{-6}	—

Key Observations:

1. **Large effect size:** Static→execution ordering achieves 29.02% relative improvement, nearly doubling our 15% threshold and substantially exceeding prior self-repair improvements.
2. **Strong statistical significance:** $p = 6.31 \times 10^{-6}$, with 95% CI excluding zero by a wide margin.
3. **Matched content control:** The 12.5 percentage point absolute improvement stems purely from presentation order.

5.2. Mechanism Analysis (RQ2)

Table 3. Early iteration gains (H-M1)

Condition	$\Delta\text{Pass}_{1 \rightarrow 2}$	p -value
Static-First (A)	12.50%	—
Exec-First (B)	12.05%	—
Difference	+0.45pp	0.859

Finding: Early gains are nearly identical. H-M1 is **not supported**.

Table 4. Regression rates (H-M2)

Condition	Regression Rate $_{1 \rightarrow 2}$	p -value
Static-First (A)	21.53%	—
Exec-First (B)	34.69%	—
Difference	13.15pp (38%)	0.0198

Finding: Static-first ordering reduces regressions by 38%. H-M2 is **supported**.

5.3. Summary

The effect operates through trajectory stabilization, not acceleration. Static-first ordering helps the LLM repair more stably, not faster.

6. Discussion

6.1. Key Findings

Ordering as a First-Order Variable: The 29% relative improvement challenges the assumption that feedback content alone determines repair quality. This effect size rivals improvements from model upgrades.

Regression Prevention, Not Acceleration: Static-first ordering prevents over-correction—aggressive changes that break previously passing tests. Clearing surface issues first creates a cleaner foundation for semantic reasoning.

6.2. Limitations

Data Source: Results use MOCK_POC execution mode. Exact magnitudes require real API confirmation.

Single Model: GPT-4o-mini only. Cross-model validation is needed.

Fixed Token Budget: 500+500 split; optimal ratio unknown.

6.3. Broader Impact

Our findings offer immediate practical value: reordering feedback is free, requires no additional compute or data, and provides substantial performance gains. We see no significant negative impacts from this research.

7. Conclusion

We began by asking whether feedback ordering affects LLM code repair quality. Our experiments provide a decisive answer: it does. Static→execution ordering achieves 29% relative improvement over execution→static, with byte-identical content in both conditions.

The mechanism is regression prevention, not early-gain acceleration. Static-first ordering doesn’t help the LLM repair faster; it helps it repair more stably—38% fewer regressions between iterations.

This finding suggests a broader principle: when designing LLM feedback systems, the structure of presentation may matter as much as the content itself. Just as human debuggers benefit from fixing typos before tackling logic errors, LLMs repair code better when feedback mirrors this coarse-to-fine structure.

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A. Appendix

A.1. Experimental Details

Full experimental logs and reproduction scripts are available in the supplementary materials.

A.2. Additional Results

Per-dataset breakdowns and iteration-level analysis are provided in the supplementary materials.